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Started	Wednesday, 21 October 2026, 11:58 AM
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Duration	3 mins 30 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The middle element is chosen randomly

Question **2**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Sort the list ✗
b. Compare the target element with the middle element in the list
c. Compare the target element with the first element in the list
d. Divide the list into two equal parts

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search requires the list to be sorted.
d. Linear search divides the list into halves.

Question **4**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is small and sorted
- c. When the list is large and sorted
- d. When the list is large and unsorted

Question **5**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Binary search
- c. Jump search
- d. Linear search ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Hash search
- b. Binary search
- c. Both (1) & (3)
- d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is sorted ✓
- c. When the list contains duplicate elements
- d. When the list is very small

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the right sublist
c. In the left sublist ✓
d. At the beginning of the list

Question **9**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is the last element in the array
c. Item is the last element in the array or item is not there at all
d. When the item is somewhere in the middle of the array ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Checking each element sequentially ✓
c. Skipping every second element
d. Dividing the list in half repeatedly

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n ✓
c. log n
d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It does not work with negative numbers

Question **13**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. log n
- c. n/2
- d. n ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search works on unsorted lists
- b. Binary search does not require dividing the list
- c. Binary search has a lower time complexity on large, sorted lists ✓
- d. Binary search can find multiple instances of the target element

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(log n)
- b. O(1) ✓
- c. O👎
- d. O(n log n)